

PAPER-1

Time : 3 Hours

Maximum Marks : 183

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **21** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble **COMPLETELY.**
12. The correct way of darkening a bubble is as :
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

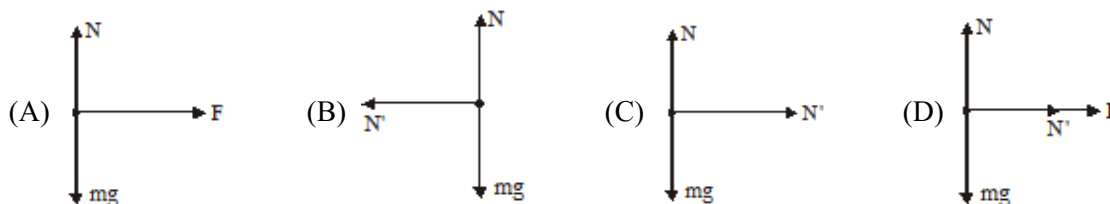
HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

PHYSICS

SEC-1 : (Maximum Marks: 28)

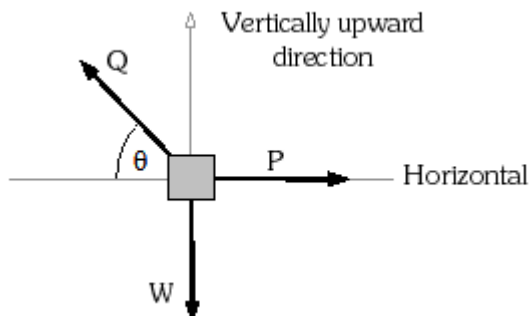
- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Choose incorrect FBD of 'A'

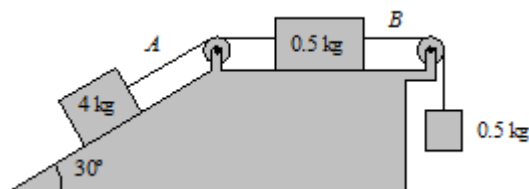


Space for Rough Work

2. A block of weight W is kept in equilibrium by two forces P and Q as shown in the figure. Which of the following give the condition for the block to be in equilibrium?



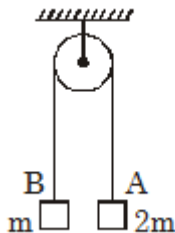
- (A) $W = P \tan \theta$ (B) $W = P \cot \theta$
 (C) $W = Q \sin \theta$ and $P = Q \cos \theta$ (D) $P = Q \sin \theta$ and $W = Q \cos \theta$
3. System shown in the figure is released from rest. Which of the following statements is/are false ?
 [$g = 10 \text{ m/s}^2$]



- (A) Magnitude of acceleration of each block is 3 m/s^2 .
 (B) Tension in the cord A is 6.5 N .
 (C) Tension in the cord B is 8 N .
 (D) Difference of magnitudes of tensions in both the cord is 1.5 N .

Space for Rough Work

4. In given diagram pulley and string are ideal, block A of mass $2m$ and B of mass m are released from rest then ($g = 10 \text{ m/s}^2$) :



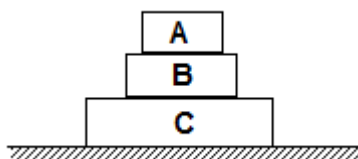
- (A) Magnitude of acceleration of A with respect to B is $\frac{2g}{3}$.
- (B) Displacement of A & B will be equal.
- (C) Distance covered by A & B in 3s will be 15 m.
- (D) Velocity of A and B are equal.
5. Equation of path of a particle moving in x-y plane is given by $y = 10x - 2x^2$ for the given particle select correct statement. Assume acceleration of gravity as 10 m/s^2 in negative y-direction :-
- (A) Maximum height of the particle is 12.5 m.
- (B) Range of the particle shown is $x = 5$.
- (C) At $x = 2.25$ particle is moving at an angle of 45° with x-axis.
- (D) Initially particle is projected at an angle of 53° .

Space for Rough Work

6. A uniform heavy rope of mass 8 kg and length 100 cm is being pulled on smooth surface as shown. Choose correct statement(s) :



- (A) Acceleration of rope will be 5 m/s^2 .
 (B) Tension in rope at 25 cm from it's left corner will be 10 N.
 (C) Tension in rope at 75 cm from it's left corner will be 30 N.
 (D) Tension in rope at 50 cm from it's left corner will be 20 N.
7. In the figure shown blocks A, B and C are kept over each other and are placed on ground then



- (A) net force on each block is zero.
 (B) normal contact force between ground and block C is equal to the combined weight of all blocks.
 (C) if an downward force is applied on block A, normal contact force on each block will increase.
 (D) Total two forces are acting on block C.

Space for Rough Work

SEC-2 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
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Negative Marks : -1 In all other cases.

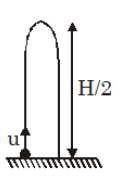
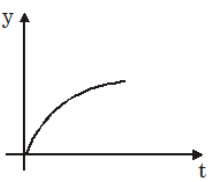
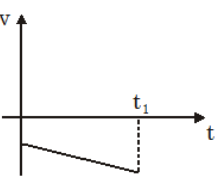
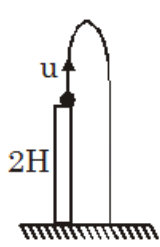
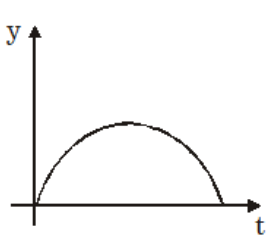
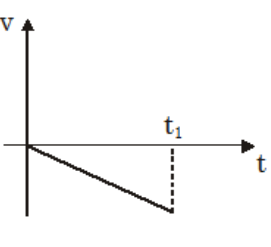
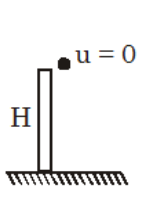
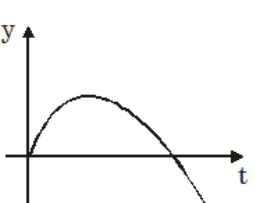
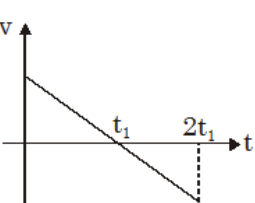
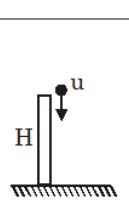
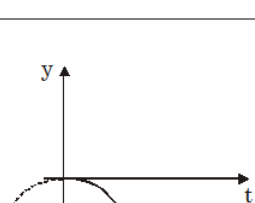
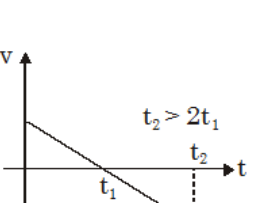
Paragraph for Questions 1 - 3

Answer Q.8, Q.9 and Q.10 by appropriately matching the information given in the three columns of the following table.

Column-1 represents different cases of vertical motion under gravity, with point of projection (or drop) as origin and upward (y) direction as positive.

Column-2 represents position-time (y-t) graph of motion

Column-3 represents velocity-time (v-t) graph of motion

	Column-1		Column-2		Column-3
(I)		(i)		(P)	
(II)		(ii)		(Q)	
(III)		(iii)		(R)	
(IV)		(iv)		(S)	

Space for Rough Work

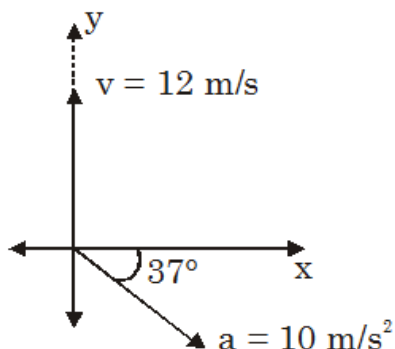
1. In which case particle is speeding down then speeding up for equal time intervals
(A) (II) (ii) (R) (B) (I) (ii) (R) (C) (II) (iii) (R) (D) (I) (iii) (S)
2. In which case particle strikes the ground with maximum speed
(A) (II) (i) (S) (B) (IV) (iii) (R) (C) (IV) (i) (R) (D) (II) (iii) (S)
3. In which case distance travelled by particle is minimum
(A) (III) (iv) (Q) (B) (I) (iv) (Q) (C) (IV) (iii) (R) (D) (IV) (iv) (P)

Space for Rough Work

Paragraph for Questions 4 - 6

Answer Q.11, Q.12 and Q.13 by appropriately matching the information given in the three columns of the following table.

Diagram shows velocity and acceleration of a particle, initially at origin at $t = 0$.



	Coloumn-1 x-coordinate		Column-2 y-coordinate		Column-3 Speed (in m/s)
(I)	0 m	(i)	0 m	(P)	10
(II)	4 m	(ii)	6 m	(Q)	16
(III)	16 m	(iii)	9 m	(R)	32
(IV)	64 m	(iv)	12 m	(S)	$4\sqrt{73}$

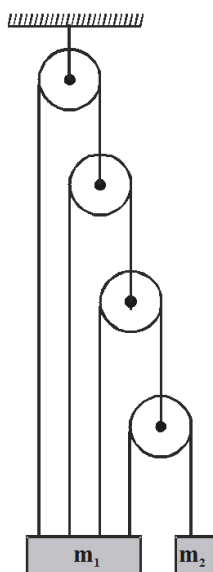
4. At $t = 1$ sec :-
 (A) (II) (iv) (P) (B) (I) (III) (Q) (C) (II) (iii) (P) (D) (III) (IV) (Q)
5. When y component of velocity is zero :-
 (A) (II) (iv) (Q) (B) (III) (iv) (Q) (C) (IV) (iii) (R) (D) (I) (iii) (Q)
6. When $v_y = -12$ m/s :-
 (A) (IV) (i) (S) (B) (I) (i) (Q) (C) (IV) (iv) (S) (D) (I) (ii) (R)

Space for Rough Work

SEC-3 : (Maximum Marks: 15)

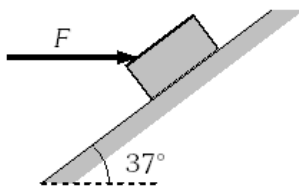
- This section contains **FIVE (05)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories:
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Zero Marks : 0 In all other cases.

1. If system is in equilibrium mass of m_1 is 15 kg. Find mass (in kg) of m_2 .

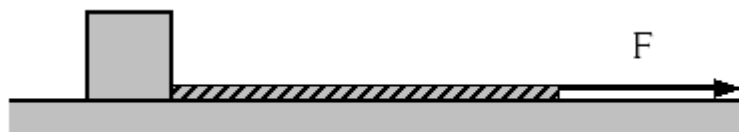


Space for Rough Work

2. A block of weight 12 N is kept at rest on a smooth slope by applying a horizontal force F as shown in the figure. Find the magnitude of the force F .

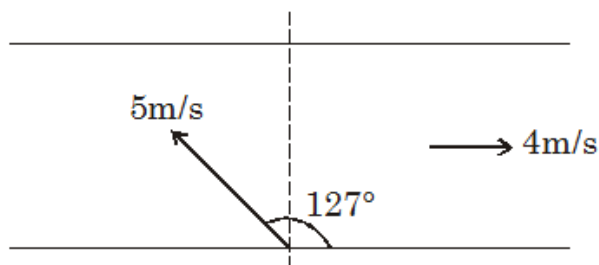


3. A uniform rope of mass 1.0 kg is connected with a box of mass 2.0 kg, which is placed on a smooth horizontal surface. The free end of the rope is pulled horizontally by a force 6 N. Find the tension in N at the midpoint of the rope.



Space for Rough Work

4. At an instant lift is moving with 10 m/s upward and with acceleration 5 m/s^2 upward. At the same instant, a coin is dropped from 7.5m above the floor of lift, find the distance travelled by coin in ground frame before it strikes the floor :-
5. In the given figure river water is flowing at 4 m/s. A boat is steered at angle 127° from the direction of flow of river. Width of river is 8 km, and velocity of boat in still water is 5 m/s. Find drift of boat in river in (km).



Space for Rough Work

CHEMISTRY

SEC-1 : (Maximum Marks: 28)

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- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

- Spin magnetic moment $\mu = 5.9$ BM is obtained -
 (A) In ground state of Mn (B) In dipositive ion of Mn
 (C) In tripositive ion of Fe (D) In uninegative ion of S
- Which of the following orbitals lobes are completely absent in nodal plane of p_y and nodal plane of p_z .
 (A) p_x (B) d_{z^2} (C) d_{yz} (D) $d_{x^2-y^2}$

Space for Rough Work

3. Which of the following process(es) is/are associated with best possibility of the energy absorbed ?
(A) $\text{Li} \rightarrow \text{Li}^+ + \text{e}^-$ (B) $\text{O}^- + \text{e}^- \rightarrow \text{O}^{2-}$
(C) $\text{Cl}^+ + \text{e}^- \rightarrow \text{Cl}$ (D) $\text{Be} + \text{e}^- \rightarrow \text{Be}^-$
4. Identify the incorrect order(s) in the following.
(A) $\text{F}^- > \text{Cl}^-$ (ionic radius) (B) $\text{Na} > \text{Mg}$ (ionization energy)
(C) $\text{O} < \text{N}$ (electron affinity) (D) $\text{F} < \text{N}$ (electronegativity)
5. Correct order of Boiling point is/are
(A) $\text{H}_2 < \text{He}$ (B) $\text{H}_2 < \text{D}_2$ (C) $\text{H}_2\text{O} < \text{D}_2\text{O}$ (D) $\text{NH}_3 < \text{SbH}_3$
6. Which of the following chemical species having permanent dipole-moment.
(A) XeF_4 (B) XeO_3F_2 (C) XeO_2F_2 (D) XeO_3
7. Which of the following species is/are planar, polar and hypervalent ?
(A) $\text{XeF}_5^\ominus$ (B) SO_2 (C) XeO_6^{4-} (D) ClF_3
-

Space for Rough Work

SEC-2 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
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Paragraph for Questions 1 - 3

Column-1	Column-2	Column-3
(I) XeF_5^+	(i) non-planar species	(P) sp^3
(II) PCl_5	(ii) Central atoms contains two lone pair of electrons on it.	(Q) sp^2
(III) ClF_3	(iii) Planar species	(R) sp^3d
(IV) CH_4	(iv) d_{z^2} orbital participate in hybridization	(S) sp^3d^2

- Which of the following is only correct combination
 (A) (I) (ii) (S) (B) (II) (i) (R) (C) (III) (iii) (Q) (D) (IV) (iv) (R)
- Which of the following is only correct combination
 (A) (I) (iii) (R) (B) (II) (iv) (S) (C) (III) (iv) (R) (D) (IV) (i) (R)
- Which of the following is only correct combination.
 (A) (I) (iv) (P) (B) (IV) (i) (P) (C) (II) (ii) (S) (D) (III) (ii) (S)

Space for Rough Work

Paragraph for Questions 4 - 6

Answer questions by appropriately matching the information given in the three columns of the following table.

Column-I Element/ion		Column-II (Magnetic behaviour)		Column-III (μ)	
(I)	Cl^-	(i)	Paramagnetic	(P)	0
(II)	Cr^{+3}	(ii)	Diamagnetic	(Q)	2.82
(III)	Cu^{+1}	(iii)	Paramagnetic	(R)	0
(IV)	S	(iv)	Diamagnetic	(S)	3.87

4. Which one of the following option is **CORRECT** combination ?
- (A) (I, ii, Q) (B) (II, ii, S) (C) (I, iii, S) (D) (II, i, S)
5. Which one of the following option is **CORRECT** combination?
- (A) (III, ii, R) (B) (IV, iv, P) (C) (III, ii, S) (D) (IV, iii, R)
6. Which are of the following option is **CORRECT** combination?
- (A) (I, ii, P) (B) (III, i, P) (C) (II, ii, R) (D) (IV, iv, P)

Space for Rough Work

SEC-3 : (Maximum Marks: 15)

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1. Number of oxy acids that can donate more than two H^+ ions in water is
 $H_3PO_3, H_2SO_4, H_3BO_3, H_4P_2O_5, H_2S_2O_7, H_3PO_2, H_3PO_4$
2. Find the total number of species having bent shape.
 $I_3^-, NO_2^+, H_2S, OCl_2, XeF_2, SO_2, ICl_2^-$
3. Find the number of identical bond angles in H_3O^+
4. Find the number of chemical species having radius less than Mg^{+2}
 $Li^+, K^+, Mg, Ca^{+2}, Sr^{+2}$
5. Find the number of subshell having $n + \ell = 7$ from the following.
 $4d, 5s, 4f, 3d, 6s, 7s, 7p, 8s, 6p$

Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 28)

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-
1. If $5^{\log_5(x^2-5x+7)} = x - 1$, then x can be -
 (A) 2 (B) 3 (C) 4 (D) 5
 2. Let $p, q, r \in \mathbb{R}^+$ and $27pqr \geq (p + q + r)^3$ and $3p + 4q + 5r = 12$, then $p^3 + q^4 + r^5$ is not equal to-
 (A) 2 (B) 3 (C) 4 (D) 5
-

Space for Rough Work

3. The equation $(\log_{10}x + 2)^3 + (\log_{10}x - 1)^3 = (2\log_{10}x + 1)^3$ has
 (A) One natural solution (B) Two rational solutions
 (C) No prime solution (D) One irrational solution
4. If 'm' is number of digits in 3^{400} and 'n' is number of zeroes after decimal and before first significant digit in 2^{-200} , (Use $\log_{10}2 = 0.3010$, $\log_{10}3 = 0.4771$) then
 (A) $m = 190$ (B) $m = 191$ (C) $n = 61$ (D) $n = 60$
5. Let $S_n = (2 + 2) + (2^2 + 5) + (2^3 + 10) + (2^4 + 17) + \dots$ upto n brackets.
 If $S_n = 2^{n+A} + Bn^3 + Cn^2 + Dn + E \forall n \in \mathbb{N}$
 (where A, B, C, D, E are constants), then which of the following is/are true ?
 (A) $A = 1$ (B) $B = \frac{1}{6}$ (C) $D = \frac{7}{6}$ (D) $E = -2$
6. Sum of infinite series $1 + \left(1 + \frac{1}{2}\right) \left(\frac{1}{3}\right) + \left(1 + \frac{1}{2} + \frac{1}{2^2}\right) \left(\frac{1}{3^2}\right) + \dots$ is less than or equal to-
 (A) $\frac{9}{5}$ (B) 10 (C) $\frac{100}{7}$ (D) 1
7. Consider an infinite geometric series with first term a and common ratio r. If its sum is 4 and the second term is $\frac{3}{4}$, then
 (A) $a = \frac{4}{7}$, $r = \frac{3}{7}$ (B) $a = 1$, $r = \frac{3}{4}$
 (C) $a = \frac{3}{2}$, $r = \frac{1}{2}$ (D) $a = 3$, $r = \frac{1}{4}$

Space for Rough Work

SEC-2 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 1 - 3

Answer questions by appropriately matching the information given in the three columns of the following table.

Column-I contains the equation with respect to x, Column-II contains sum of their roots and Column-III contains product of their roots.

Column-I		Column-II		Column-III	
(I)	$x^2 - 3(5^{\log_5 x}) + 2 = 0$	(i)	3	(P)	-12
(II)	$\log_3(x^2 - 2) + \log_3(x - 1) = \log_3 2$	(ii)	2	(Q)	1
(III)	$ x - 2 ^{(x^2 - 4)} = 1$	(iii)	$\frac{17}{4}$	(R)	-6
(IV)	$2^{(\log_2 x)^2} + x^{\log_2 x} = 32$	(iv)	4	(S)	2

- Which of the following options is the only **CORRECT** combination ?
 (A) (I) (ii) (R) (B) (III) (iii) (P) (C) (II) (i) (P) (D) (II) (ii) (S)
- Which of the following options is the only **CORRECT** combination ?
 (A) (I) (i) (P) (B) (III) (iv) (P) (C) (II) (i) (R) (D) (IV) (iii) (Q)
- Which of the following options is the only **CORRECT** combination ?
 (A) (I) (ii) (S) (B) (I) (i) (S) (C) (II) (ii) (Q) (D) (III) (iv) (P)

Space for Rough Work

Paragraph for Questions 4 - 6

Column-I : Contains inequations.

Column-II : Contains complete solution set of inequations of Column-I.

Column-III : Contains number of integral solution of inequations of Column-I.

Column-I		Column-II		Column-III	
I	$\frac{(x-1)(x-2)}{x-3} \geq 0$	(i)	$(-2, -1) \cup (0, 5) - \{2\}$	(P)	0
II	$\frac{x}{x+1} > 2$	(ii)	$[1, 2] \cup (3, \infty)$	(Q)	1
III	$\frac{x^2 + 4x + 4}{2x^2 - x - 1} < 0$	(iii)	$(-2, -1)$	(R)	finite & more than 1
IV	$\frac{x^3(x-2)(5-x)}{(x^2-4)(x+1)} > 0$	(iv)	$\left(-\frac{1}{2}, 1\right)$	(S)	infinitely many

4. Which of the following options is only the **CORRECT** combination ?
 (A) (I) (ii) (S) (B) (II) (iii) (Q) (C) (I) (i) (R) (D) (III) (iv) (S)
5. Which of the following options is only the **INCORRECT** combination ?
 (A) (II) (iii) (P) (B) (III) (iv) (Q) (C) (I) (iii) (Q) (D) (IV) (i) (R)
6. Which of the following options is only the **CORRECT** combination ?
 (A) (III) (iv) (P) (B) (II) (iii) (P) (C) (IV) (i) (S) (D) (I) (ii) (R)

Space for Rough Work

SEC-3 : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories:

Full Marks : +3 If ONLY the correct answer is chosen.

Zero Marks : 0 In all other cases.

1. If a, b, c are positive real numbers such that $ab^3c^4 = 256$, then the least value of $\left(\frac{1}{a} + \frac{3}{b} + \frac{4}{c}\right)$ is
2. If $S = 3.\sqrt{9}.\sqrt{\sqrt{27}}.\sqrt{\sqrt{\sqrt{81}}}\dots\infty$, then sum of digits in S is
3. If 11 arithmetic means $A_1, A_2, A_3, \dots, A_{10}, A_{11}$ are inserted between 10 and 28, then value of $(A_8 - A_4)$ is equal to
4. The number of solutions of the equation $\sqrt{\log_{10}(x^2 + 7)} = \log_{(x^2 + 7)}\left(\frac{1}{10}\right)$ is
5. The value of $\log_{\sqrt{2}}\left(\text{antilog}_{128}\left(\frac{3}{7}\right)\right) + \log_2\left(\text{antilog}_{128}\left(\frac{3}{7}\right)\right)$ is equal to

Space for Rough Work

Space for Rough Work